ICT Unit A: Information Processing

Ch1 Introduction of Information Processing

➢Input-Process-Output Cycle

- Input (data of attendance) → Process (search for absence list) → Output (attendance report)

➢Stored program

- Program: A sequence of instructions → Instruct the computer → Complete 1 specific task per time
- Stored program: Inside ROM (Read Only Memory)

➢ Data vs Information

- Data (Collection of raw fact + Not organized) → Processed → Information

➢ Data types

- Text: characters stored in binary codes, e.g. Unicode
- Images: Colour-bit represented by binary code, e.g. Bitmap
- Audio: Digital signal (Conversion of analogue signal), e.g. WAV
- Video: Sequence of images, e.g. MPEG

➢Components of Information System (For data processing)

1. Propose:
   - Automate existing manual process
   - Improve existing system
   - Fix system problems
2. Data:
   - Maybe stored in database (structured collection of records)
3. Process:
   - Stored data processing
4. Technology:
   - Hardware: Physical components
   - Software: System software and application software
   - Network: Computers connection
5. Personnel:
   - End users
   - IS specialist (e.g. Network manager, Programmers, System analysts): Develop + Manage

➢ Information life cycle

- Data collection → Organization → Storage → Processing → Analysis → Transmission Presentation
- **Knowledge-based society**: A society that deploys Knowledge + IT + Communication System

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Ch2 Data Organization and Data Control

Concept.1 File access mode

- Sequential access mode: Access data bit by bit, e.g. Magnetic tape
- Direct access mode (8 ~ 20 ms seek time) : Access data at anywhere randomly, e.g. Hard disk

Concept.2 Data

- Data error source:
  1. Data source error
  2. Transcription error (人為的抄寫錯誤)
  3. Transposition error (accidentally reverses two adjacent digits)
- GIGO: Erroneous (錯誤) data → Garbage-In-Garbage-Out → Inaccurate information

➢Data control

1. Data validation:
   - **Field Presence** check (e.g. Must-fill fields)
   - **Field Length** check (e.g. 10 digits for address)
   - **Range** check (e.g. Height of adults)
   - **Type** check (e.g. Numeric data for age)
   - **Fixed Value** check (e.g. T/F , M/F)
   - **Format** check (e.g. Email with '…@...')
   - **Check digit** - Self-checking: **Parity Check** (e.g. ISBN, HKID Card)

Check digit example (for ISBN & HKID): 962-943-717-1

⇒ 9x10 + 6x9 + 2x8 + 9x7 + … + 7x2 + 1x1 = 308, which is divisible by 11 ⇒ a valid ISBN

*It is same for check digit of HKID number, the sum need to be **divisible by 11**

2. Data verification:
   - **Input data twice**
   - **Double data entry**

➢Hierarchy (等級制度) of database

- Field (Specific fact) < Record (Information of an entity) < Table (Collection of entities) < Database

➢ DBMS – Database Management System (software)

⇒ Manage database structure + Storage + Organization + Retrieval of data

⇒ Establish **relationships** among tables (Better than file manager)

- E.g. MySQL (Database server). MS Access (Small office, personal use)
- Record filter: Unmatched records are not accessible temporarily ← Filter constraints
- Sorting:
  - **Sorting function**: Save sorted records in a new table
  - **Database index**: Specify index expression with **index command** → Arrange order
  - Small file of index key → Load index into the RAM
- Other elements in DBMS:
  - **Data entry forms**: UI for ADD + MODIFY of records
  - **Query forms**: UI for EXTRACT + DISPLAY of records
  - **Reports**: Display / Print results in defined format

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Ch3 Number & Character Coding Systems

Concept.1 Number systems

1. **Denary number system** → Counting + Calculation
2. **Binary number system** → Data representation + Manipulation in computers
3. **Hexadecimal number system** → Facilitate communication between computers and programmers

➢ Conservation of number systems

- For Hex number, 1 digit (Hex. Number) = 4 bit (binary number) , e.g. $F_{16} = 1111_2$

Concept.2 Number representation in computer systems

- 8 bit (Basic unit of binary digit) = 1 Byte (Smallest addressable unit in microprocessor) < 1KB (= 1024B)
- **Word**: Natural unit of data used by computer
- **Word length**: e.g. 16 bits, 32 bits, 64 bit
- **Representation of negative numbers:**
  1. **Sign-and-magnitude** : Sign the leftmost bit (1xxx = negative ; 0xxx = positive)
  2. **Two's complement**: i. **One's complement**: Invert all digits (from 1 to 0 / from 0 to 1)
     ii. One's complement + 1

n-bit number representation	Smallest integers can be represented	Largest integers can be represented
Unsigned	0	$2^n - 1$
Sign-and-magnitude	$-(2^{n-1} - 1)$	$(2^{n-1} - 1)$
Two's complement	-2^{n-1}	

- Overflow error (Occur when the number cannot be represented by fixed number of digits)
  ← Addition of 2 numbers of the same sign (±) + Opposite sign of the result (∓)

➢Character coding systems

Coding system	Features	Length of digits
ASCII (American Standard Code for Information Interchange)	Alphanumeric characters (A : 65, a : 97) + Graphics symbol	7 bits
Big5 code	For Traditional Chinese	2 bytes
Guo Biao code (GB code)	For Simplified Chinese	
Unicode	Represent all the existing languages in the world	1 ~ 4 bytes

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Ch4 Multimedia Elements and Digitization

Concept.1 Digital multimedia elements

➢ Image / Graphics

- **Bitmap Image** – Collection of colour dots (Pixels)
  - **Colour depth (bit)**: Number of bits to represent the colour of a pixel (e.g. 24 bit true colour)
  - **Resolution**: Number of pixels in 2 dimensions
  - **Printing resolution**: Measured in ppi (pixels per inch), dpi (Scanned image), dot pitch (Monitor)
  - E.g. bmp (Bit Map), jpeg (Joint Photographer Expert Group), png (Portable Network Graphics)
- **Vector Graphics** – Composition of several objects (Shape, line, string or text)
  - **Shape + Location of vector objects**: Defined by **Mathematical formulae**
  - **Object Properties**: Fill colour, thickness of outline, order of layer (like PS)
  - E.g. wmf (Windo Metafile), eps (Encapsulated Postscript), ttf (True Type Font)

➢ Audio – Digitized analogue signal (Data of longitudinal wave)

- **Bit resolution (bit)**: Number of bits Represent the amplitude of an audio signal
- **Sampling rate (Hz)**: Number of samples taken per second (During conversion)
- **Stereo audio**: 2 streams of audio signals (Better listening experience)
- E.g. wav (Wave), ogg, mp3

➢ Video – Sequence of full-motion images + Sound tracks

- File size ∝ Frame rate + Frame size + Duration + Compression method
- E.g. mov (Quick Time Movie), avi, wmv (Windows Media Video)

Concept.2 Analogue Data vs Digital Data

Characteristics	Analogue data	Digital data
Measurement	Continuous signal	Discrete form in binary number
Access mode	Sequential Access Mode	Direct Access Mode
Information Accuracy (e.g. Data loss rate)	Dependent to sampling technique	Low
Analysis Efficiency (e.g. Search for a word)	Low	High
Transmission Efficiency	High transmission error rate	Low transmission error rate
Replication (Copy)	Impossible for absolute replication	Error free + Accurate replication
Storage Efficiency	Low	High

➢ Digitization – Analogue Data → Digital Data

Procedures	Method	Example
1. **Sampling**	Obtain discrete sample data / time interval	2 sample data per second = 2Hz
2. **Quantization**	Attribute each sample data → Digital code	400 dB to 100 dB

➢ Common multi-media elements

Elements	Method / Features	File extension
Audio	**Silicon Chips** (Analogue Digital Convertor + Digital Analogue Convertor)	Mp3, WAV, ogg
Images	Colour mode: Halftone (16bit/pixel) < Grey-scale (8bit/pixel)	Bmp, png, JPEG
Video	VCR (Video Cassette Recorder), TV Tuner card, Charge-Coupled Device	AVI, mov, Mp4

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Ch5 Introduction to Office Automation

➢ Definition

- **Office automation (Mode of office operation)**

  ⇒ Utilizes computer Hardware & Software Manipulate → digital information → Accomplish tasks

  ⇒ Productivity↑ + Data management↑ + Communication↑ + Company expense↓

➢ Hardware & Software

- Application Software: e.g. Word processor, DBMS software
- Hardware: e.g. Printer, Network Interface card (NIC card)

➢ Basic activities

1. **Data storage**
   - By secondary storage device
2. **Data management**
   - **DBMS**: Efficient data access + Report generation
   - **Task scheduling system:** At timeline manage resources and use activities (Monitor + Manage) + Resources (Manipulate) + Timeline , e.g. Gantt Chart generation
3. **Data exchange**
   - Through electronic communication channels
   - E.g. IM, groupware (Access data in central repository 知識庫), video conferencing
4. **Data application**
   i. Data analysis
   ii. Digital Image Modification
   iii. Document Production
   iv. Report Generation

➢ Effectiveness

- **Determinants**: Installation + Maintenance Cost, **Compatibility** with existing system, Implementation time, Levels of **technical** equipment, Physical arrangement of equipment

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Ch6 Word Processing

➢ Word processor

⇒ Create, edit, format, save, print document

Keywords	Definition	Examples
File format		Pdf, txt, rtf, doc
Saving	Copy document from Main Memory → Secondary Storage Device	
Printing	Send a document to printer on a medium	
Text Editing	Modify the content of a document	
Formatting	Polish appearance of a document(能看見的改變)	
AutoCorrect	Automatically change characters into others	:) →☺
Auto Format	Automatically change to predefined style	

- Others: Table of content, Index, Footnotes, Header, Paste Special, Mail Merge
  - Mail Merge contains: Merge field, Connect tables / Database / Spreadsheet
- Template > Theme (Text color, margins, page borders) > Style (Predefined characters, paragraph format)

Ch7 Spreadsheet

➢ Electronic Spreadsheet

⇒ Input, organize, create charts, analyse data

⇒ Applications: Generate Report, Mailing list

⇒ E.g. Microsoft Office Excel, LibreOffice.Calc

➢ Cells and cell reference

- Cells < Table < Spreadsheet
  - With unique cell address
  - Contain data or formulae
  - **Cell format**: General, Currency, Date, Time, Fraction, Scientific, Fixed value
  - **Autofill**: Fill up selected cells with **Specified repeated pattern / Extrapolation** 推算

➢ Functions

Elements	Description
Constant	Raw Data
Operator	• Arithmetic operator: + - * / ^ % • Comparison operator: = <> > >= < <= • Text concatenation operator: &
Cell reference	• **Relative Reference**: Column + Row Intercepts , e.g. A2 • **Absolute Reference**: Absolute column + Absolute row , e.g. $A:2 • **Mixed Reference**: Mix with the above , e.g. $A2 , A$2
Function	Predefined formula

Statistical Functions

→ Find out tendency, distribution, characteristics

Formula	Description
=MAX(A2:A6) **=MAX(A2:A6, 30)**	Largest value in the range A2:A6 Largest value in the range A2:A6 and the value 30.
=MIN(A2:A6) **=MIN(A2:A6, 0)**	Smallest of the numbers in the range A2:A6 Smallest of the numbers in the range A2:A6 and 0
=LARGE(A2:B6 ,3)	3rd largest number in the numbers of the range
=SMALL(A2:A10, 4)	4th smallest number in the number of the range
=AVERAGE(A2:A6) **=AVERAGE(A2:A6, 5)**	Average of the numbers in cells A2 through A6 Average of the numbers in cells A2 through A6 and the number 5
=COUNT(A2:A7)	Counts the number of cells that contain numbers in cells A2 through A7
=COUNTIF(A2:A5,"apples") **=COUNTIF(B2:B5,">55")**	Counts the number of cells with 'apples' in cells A2 through A5 Counts the number of cells with a value greater than 55 in cells B2 through B5
=MODE(A2:A7)	Mode, or most frequently occurring number of the range

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Mathematical Functions

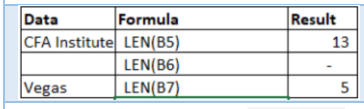
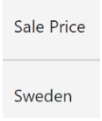
→ Manipulate numbers

Formula	Description
=SUM(A2:A10)	Adds the values in cells A2:10
=SUM(A2:A10, C2:C10)	Adds the values in cells A2:10, as well as cells C2:C10
=SUMIF(A2:A5,">160000",B2:B5)	Sum of the commissions for property values over $160,000
=SUMIF(A2:A5,">160000")	Sum of the property values over $160,000
=SUMIF(A2:A5,300000,B2:B5)	Sum of the commissions for property values equal to $300,000
=SUMIF(A2:A5,">" & C2,B2:B5) 	Sum of the commissions for property values greater than the value in C2
=ROUND(2.15, 1)	Rounds 2.15 to one decimal place
=INT(8.9)	Rounds 8.9 down
=MOD(3, 2)	3/2 的餘數 (Remainder)
=POWER(5,3)	5^3
=RAND()	A random number >= 0 and < 1
=RAND()*100	A random number >= to 0 and < 100

Logical Functions

Formula	Description
IF(<condition>, value_for_true, value_for_false) **=IF(C2="Yes",1,2)**	IF(C2 = Yes, then return a 1, otherwise return a 2)
=NOT(A2>100)	A2 is NOT greater than 100
=AND(A2>1,A2<100)	Displays TRUE if A2 is greater than 1 AND less than 100, otherwise it displays FALSE
=OR(A2>1,A2<100)	Displays TRUE if A2 is greater than 1 OR less than 100, otherwise it displays FALSE

Text Functions

Formula	Description
=CODE("A")	Displays the numeric code for A which is 65
=CHAR(65)	Displays the character represented by 65 in the computer's character set which is A
	LEN show the number of characters in a cell Spaces are counted as characters
=LEFT(A2,4) **=RIGHT(A2,5)** 	First four characters in the first string which is = sale Last 5 characters of the first string which is = price
=MID("Fluid Flow",7,4)	Starting at the 7th, 4 characters from the string which is = Flow

Data	Formula	Result
CFA Institute	LEN(B5)	13
	LEN(B6)	-
Vegas	LEN(B7)	5

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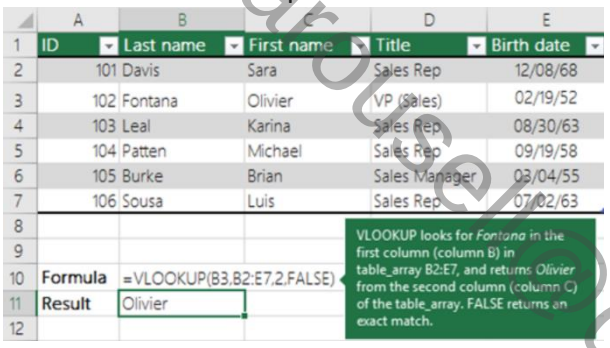
• Date / Time Function

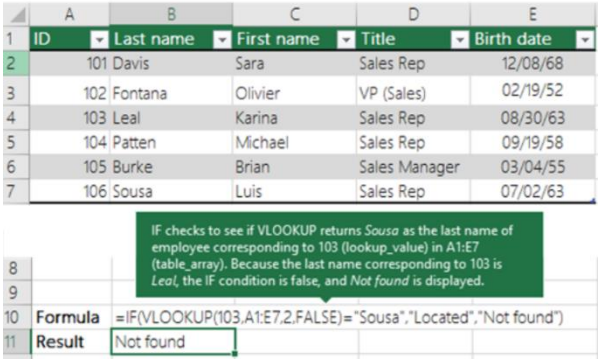
- DAY(<date>), MONTH(<date>), YEAR(<date>), WEAKDAY(<date>), TODAY(), TIME(hr, min, second)

HLOOKUP, VLOOKUP Functions

Function	Description	Example
HLOOKUP (value_to_search_in_1st_row, cell_range, column_number, not_exact_match)	not_exact_match: FALSE → find an **exact** match TRUE → find an **approximate** match (If an exact match is not found, then the HLOOKUP function will look for the next largest value than is less than value)	=HLOOKUP(10251, A1:K3, 2, False) Looks up "10251" in row 1, and returns the value from row 2 that's in the same column
VLOOKUP (value_to_search_in_1st_column, cell_range, row_number, not_exact_match)		=VLOOKUP(10248, A1:B21, 2, True)

Some VLOOKUP examples:





➢ Charts – Visualize the relationship between data

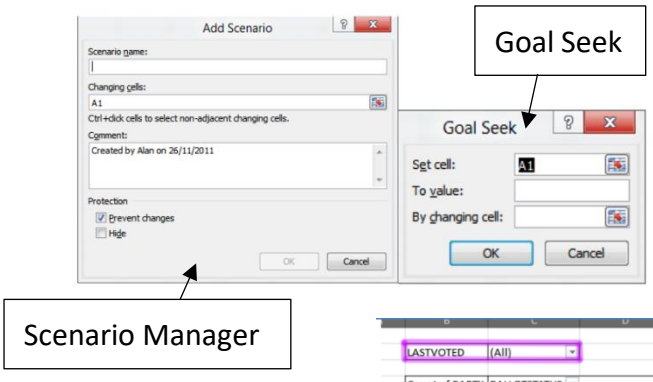
- E.g. Line chart, pie chart, bar chart, scatter chart, …

➢ Data manipulation technique (Filter function)

- **Sorting (Filter function)**: Simple sort list / Sorting with criteria → Reorder the records
- **Filtering (Filter function)**: Filter out those don't meet the specified criteria

➢ Data analysis

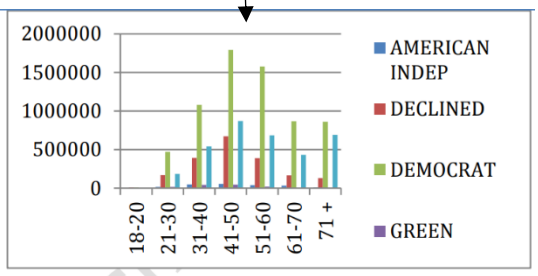
1. What-if analysis	i. Scenario Manager	Create several **input value set** → Compare outcome
	ii. Goal Seek	**Change the values** of variables → Find effects
2. Pivot table		Create summary table of a set of data
3. Pivot chart		Graphical presentation of pivot table



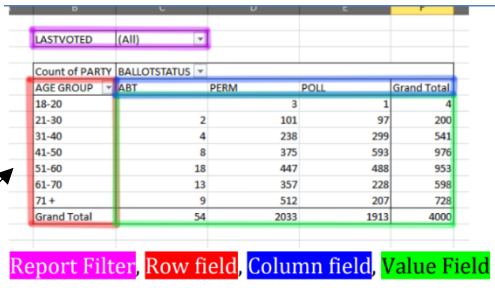
Goal Seek

Scenario Manager

Pivot Chart



Pivot Table



Report Filter, Row field, Column field, Value Field

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Ch8 Database

➢ Definition

- **Database**: Collection of data related to a particular topic / purpose

➢ DBMS – Data field

- **Field settings**: Field Name + Data Type + Field Length
- **Key field**: Unique in specified record
- **Input mask**: Prevent users from inputting invalid data

➢ DBMS – Data Type (Microsoft Access)

	Data type	Description
Typical	Text	Max 2^8 – 1 character (1 byte – 1)
	Memo	Max 2^{16} – 1 character (2 bytes – 1)
	Number	Numeric entry
Special	Date / Time	-
	Currency	-
Predefined	AutoNumber	Automatically assigned by Access (Once a new record is created)
	Yes / No	Yes / No , True / False , On / Off
External Files	OLE Object	Object Linking and Embedding objects
	Hyperlink	-

➢ DBMS – Database objects (Microsoft Access)

Database Objects	Description
Table	Row (Horizontal) x Column (Vertical)
Form	• GUI (Graphical User Interface)→ Manipulate + Display data • Form Wizard: Facilitate form creation
Report	Summarize information: Professional + Easy-to-read layout
Query	Retrieve data from different tables (Calculation + Reorder data)
Macro (Program)	→ Task automation
Module (Program)	With **data management logics** defined by users

- **Datasheet View**: Display **a table** organized in rows and columns
- **Design View**: Display **underlying structure (Field settings)** of a table

➢ DBMS – Queries SQL (Structured Query Language)

Common tasks		SQL Commands example
Table	Add	**CREATE TABLE** student (Name char(24), ID char(8), Age decimal(3, 0))
	Modify	**ALTER TABLE** student **DROP** Age
	Delete	**DROP TABLE** student
Records	Add	**INSERT TABLE** student **VALUE** ('Peter','1244',16)
	Modify	**UPDATE TABLE** student **SET** (Age = Age + 1) WHERE (Age >= 16)
	Delete	**DELETE FROM TABLE** student **WHERE** Age <14
Selection		**SELECT** field name **FROM** table name **WHERE** criteria **ORDER BY** field name, **DESC GROUP BY** field name
		DESC = descending ; **ASC** = ascending

Creation Methods: Query Wizard, Design View (GUI tool), SQL View (Text-input interface + SQL statement)

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Ch9 Integrated Use of a Software Suite

➢ Need for integration of different data

⇒ Incorporate (combine) different types of multimedia elements

➢ Methods for data integration

Methods		Explanation
Data Import		Import compatible object to the destination document
OLE	Object Linking	Create linked object that links to an actual object
	Object Embedding	Copy an object to destination document

➢ OLE – Definition + Application

- **OLE Technology** (devised by Microsoft): Sharing data among different Microsoft Software
- **OLE Object**: Data + Application → Modify data
- **Software suite**: A group of software designed to work together (e.g. Microsoft Office, Flash Player)
- Application – **Mail Merge**:
  - Word processor → Integrate structured data source (e.g. Database file) into a document template
    ⇒ Produce multiple documents (e.g. letters, email)

Ch10 Presentation of Information

➢ Ways of presentation

1. List	**Point Form**	**• Ordered List, Unordered List, Nested List → Emphasize key points**
2. Table	Tabular form	• List records + Present information orderly + Compare features of items
3. Chart	Graphical data presentation	• Enhances reader's understanding • Facilitate data analysis • Interpret results
4. Graphical Presentation	Means of symbols	• Arouses 喚起 interests of audience • Conveys 傳達 messages • Present information in organized manner

➢ Presentation creation process

1. Planning of the presentation
   - Storyboard
   - Topic + Content + Form + Means of presentation
2. Collection + Selection of appropriate information
3. Organization of relevant information
   - Introduction → Body → Conclusion
4. Introduction of interactive elements
5. Conduction of the presentation

- **Storyboard**: Visual plan for presentation → Sketches the outline of the Content, Navigation + Design elements (e.g. Tone, layout, BG Colour, Text attribute, Text formatting)
- Means of presentation:
  - Slide show, Web-based presentation (e.g. Remote slide show), Multimedia document (Flash Video)

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